

The Model 686: Berkeley Nucleonics' 20 GS/s, 14-Bit Arbitrary Waveform Generator

Two or four synchronized analog channels, up to 32 digital lines, and a synchronization bus that scales to 16 analog and 128 digital channels.

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SAN RAFAEL, Calif., May 25, 2026. When it comes to cutting-edge signal generation, precision and performance are non-negotiable. Berkeley Nucleonics Corporation (BNC) is highlighting the Model 686 arbitrary waveform generator (AWG), the fastest 14-bit AWG in the company's lineup, for engineers evaluating high-speed signal generation across research, defense, industrial, and academic test applications.

The Model 686 pairs a 20 GS/s real-time sample rate with true 14-bit vertical resolution — a combination that lets engineers generate high-fidelity signals without trading accuracy for speed, or the reverse. Multiple synchronized analog and digital channels, a scalable multi-instrument architecture, and deep per-channel memory round out an instrument built to grow with a test program rather than be outgrown by one.

Key Features and Specifications

Performance

- Sample rate programmable from 1 S/s to 20 GS/s
- True 14-bit vertical resolution for signal accuracy and detail
- Measured 3 dB bandwidth of 5.8 GHz (sin(x)/x compensated)

Signal generation

- 2 or 4 fully synchronized analog channels
- Up to 32 digital lines for mixed-signal generation
- Square-wave rise/fall time of 120 ps (10% to 90%)
- Output amplitude up to 5 Vpp (to 3.5 GHz), tapering to 3 Vpp at 6.5 GHz

Scalable design

- Available in 2- or 4-channel configurations
- Dedicated intra-chassis synchronization bus links up to 4 units for a 16-analog, 128-digital-channel system

Memory

- Up to 9 Gpts of waveform memory per channel in Full Rate mode

Software

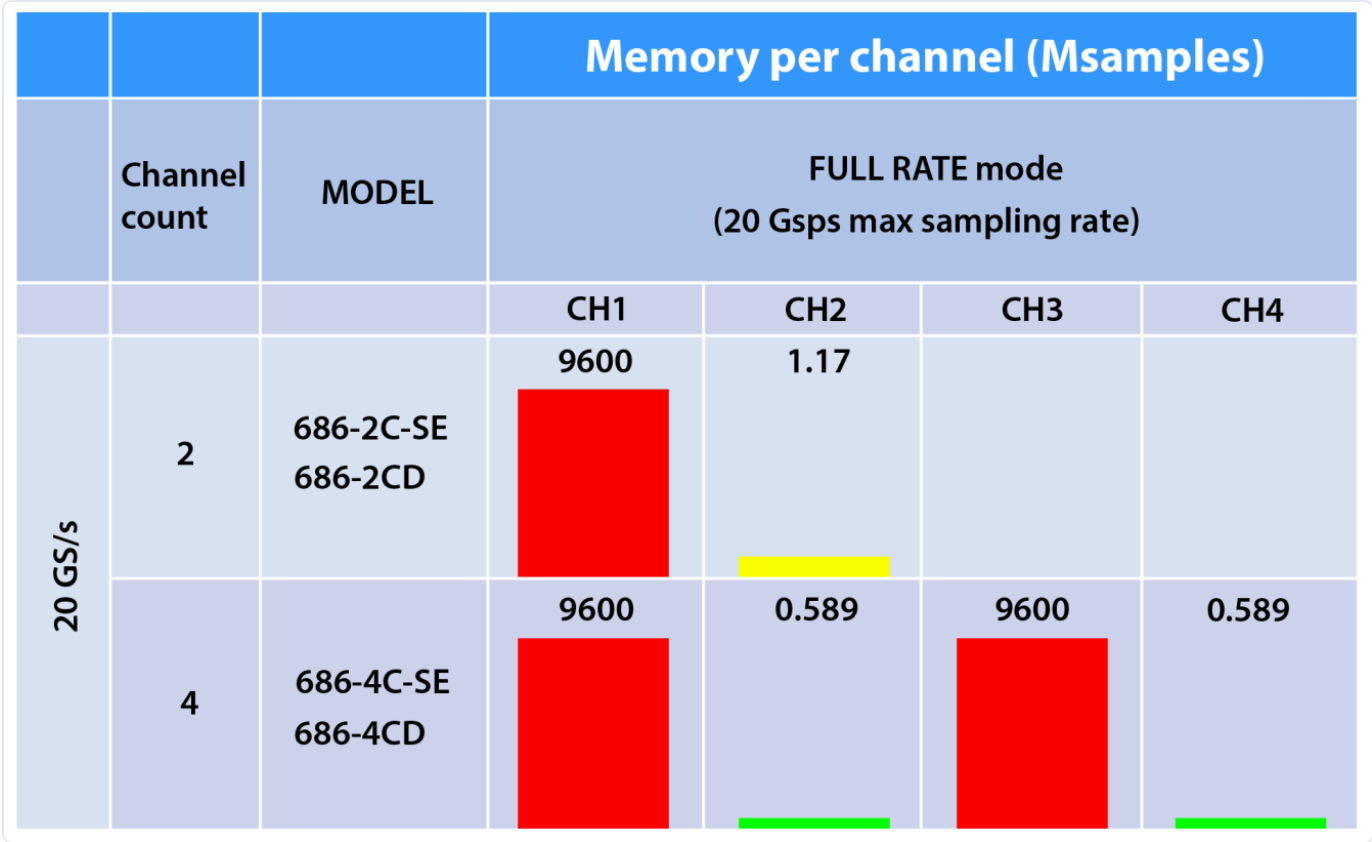
- Simple Rider AFG mode (DDS function generator)
- True Arb mode (variable-clock arbitrary waveform)
- Optional PPG mode: PRBS and multi-level PAM patterns, custom patterns up to 12 MSymbols, pattern rates up to 6.5 Gbaud
- Touchscreen-ready interface across all three modes

Applications

The Model 686's capabilities extend across a wide range of fields, including:

- Semiconductor test and device characterization
- Quantum key distribution (QKD) and quantum sensing
- RF and wireless communications
- Aerospace and defense applications
- Optics and photonics research

“The Model 686 is the instrument I point customers to when they need serious speed and resolution without picking between the two,” said David Gil, Technical Sales Representative at Berkeley Nucleonics. “It shows up on test benches solving very different problems — semiconductor validation one week, a photonics lab the next — because 20 GS/s at 14 bits is useful anywhere signal fidelity matters.”



Waveform memory per channel (Msamples) by model, Full Rate mode at 20 GS/s.

Availability

The Model 686 is available through Berkeley Nucleonics and its representatives worldwide. Full specifications, configuration options, and quote requests are on the [Model 686 datasheet page](#).

About Berkeley Nucleonics

Berkeley Nucleonics Corporation (BNC) has been a trusted name in precision electronic instrumentation since 1963, serving defense, research, academic, and industrial customers with digital delay generators, arbitrary waveform generators, RF/microwave signal generators, pulsed

power systems, and nuclear radiation detection equipment. With 60+ years of reliability and responsive technical support, BNC is flexible and listens to your application.

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